Oracle Utilities Customer Care and Billing v26.4HOT

Product Fix Design

Bug 39003305

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Oracle Customer Care and Billing Product Fix Design

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Reported Issue

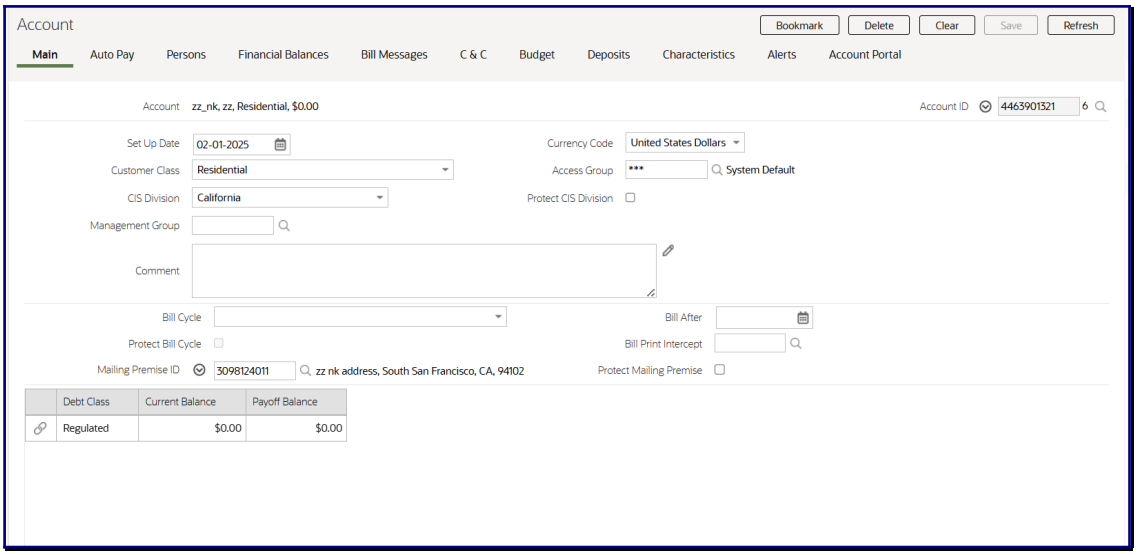
Description

A system error occurs when billing for February in a non-leap year if Service Quantity based Calculation Rule is configured with a seasonal period from February 1 to February 29.

Steps To Recreate

Data Setup

- Create a V Model.



Account

- Create a Calculation Group with Calculation Rule.

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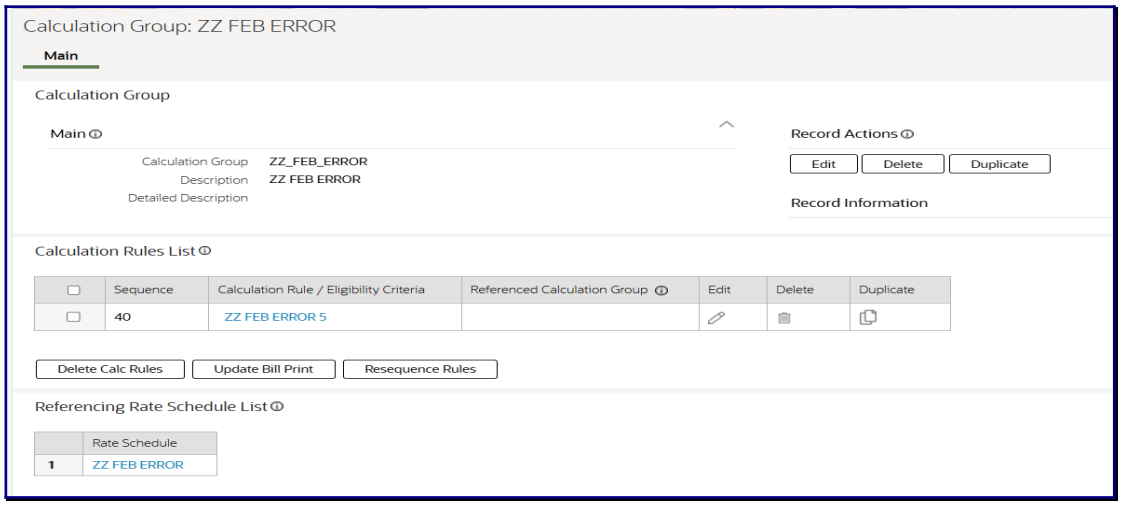
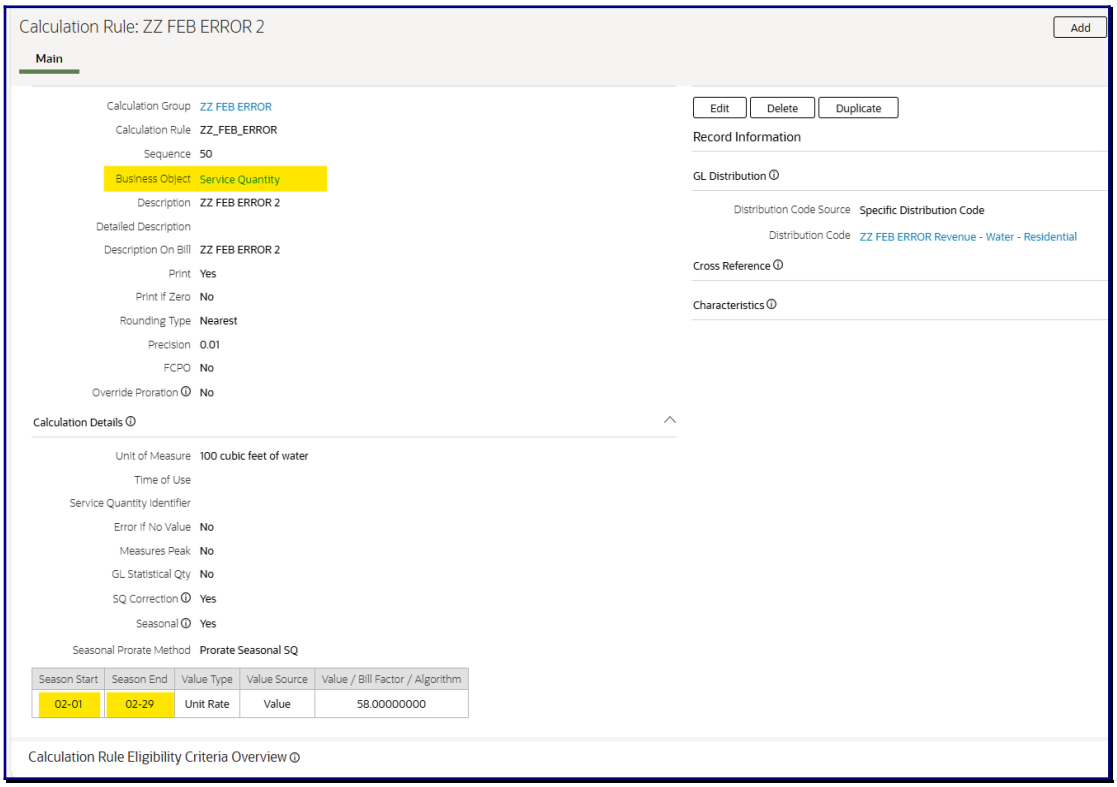


Calculation Group

- Create a Calculation Rule with Service Quantity Business Object and along with seasonal period configured.
- Provide Season Start date as 02-01 and Season End date as 02-29.



Calculation Rule

- Create a Rate Schedule with above created Calculation Group.

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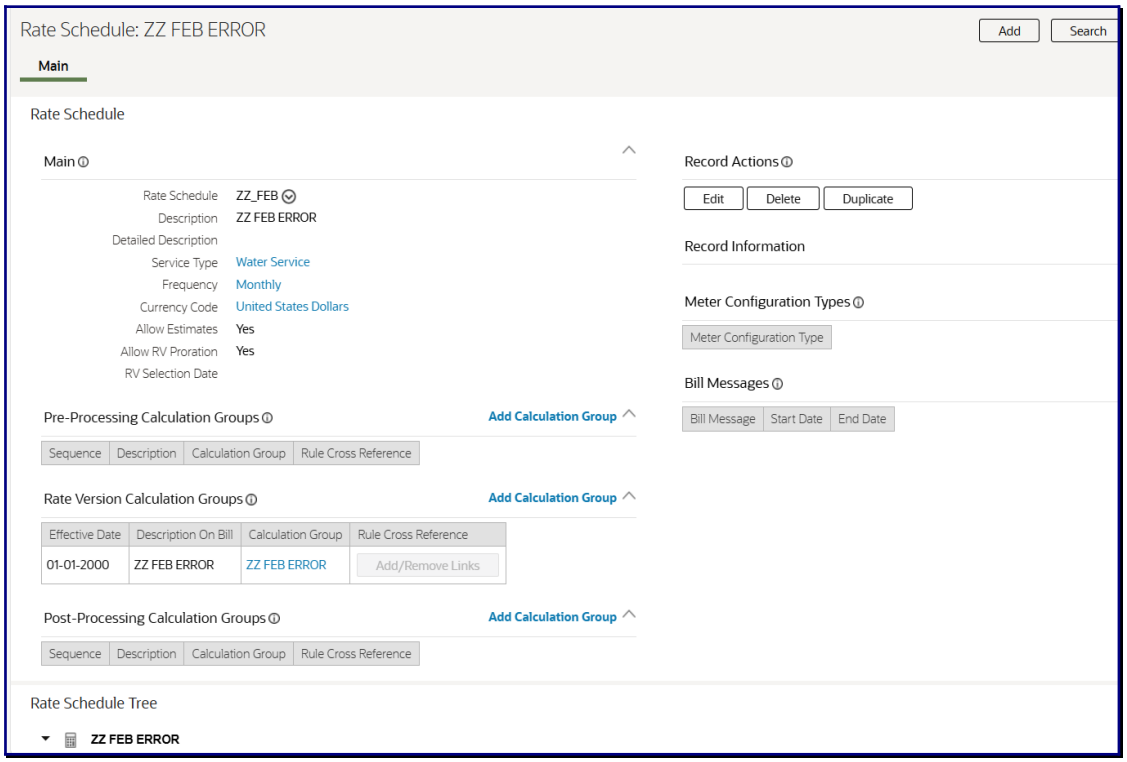
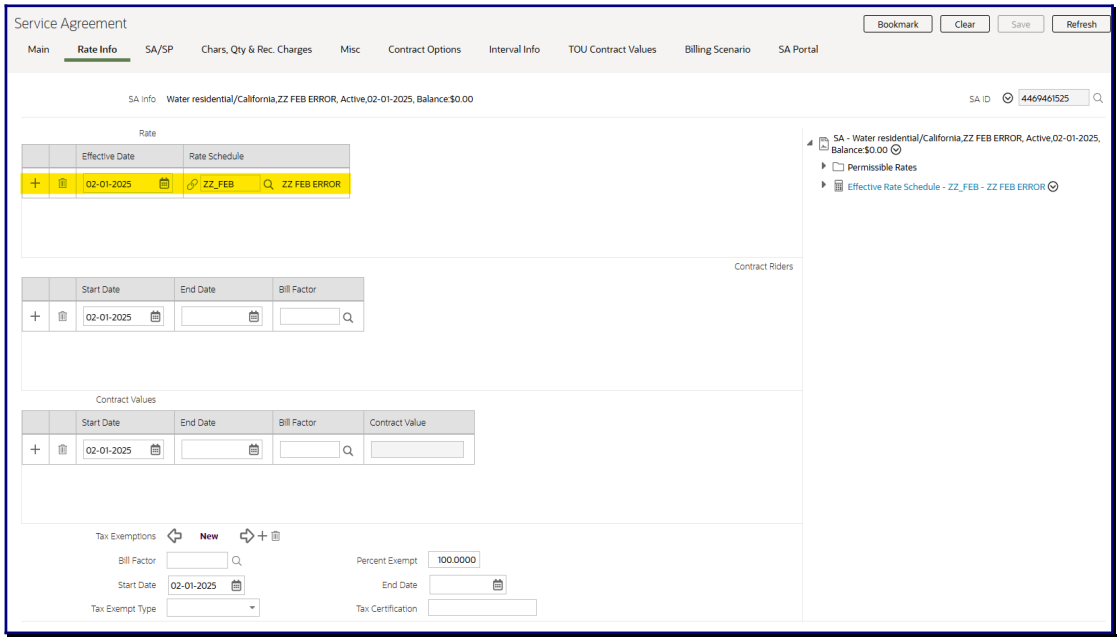


Rate Schedule

- Keep the above created Rate Schedule in SA Rate Info tab.



Service Agreement

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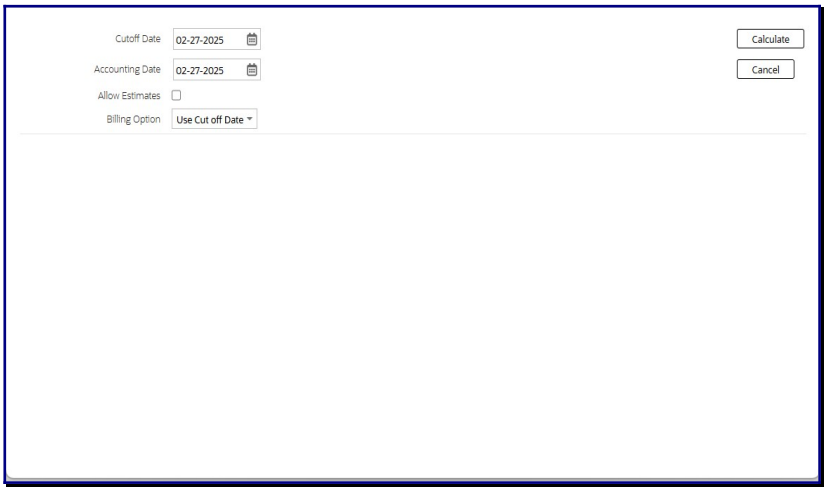
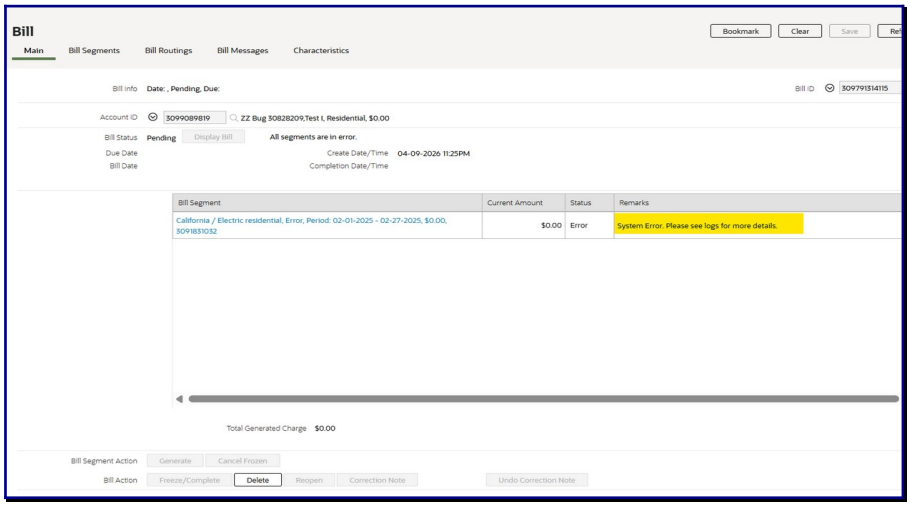
Reported Issue

- Generate a bill for period 02-01-2025 to 02-27-2025.



Bill

- "**System Error**" is encountered.



Bill

- **"Illegal Argument Exception"** error is encountered in logs.

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```
    at weblogic.work.ExecuteThread.run(ExecuteThread.java:370)
Caused by: java.lang.IllegalArgumentException: Unable to create Date 2025/2/29
    at com.splwg.base.api.datatypes.Date.ensureValidFields(Date.java:162)
    at com.splwg.base.api.datatypes.Date.<init>(Date.java:106)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.updateBillCalculationLineValuePeriod(ApplyRateToSQ_Impl.java:1165)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.updateBillCalculationLineWithForExtraFields(ApplyRateToSQ_Impl.java:1
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.prepareCalcLine(ApplyRateToSQ_Impl.java:1101)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.prorateSQAndValue(ApplyRateToSQ_Impl.java:448)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.deriveValue(ApplyRateToSQ_Impl.java:1636)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl.applyRateToSQ(ApplyRateToSQ_Impl.java:423)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl$ByteBuddy$leA7obiG.applyRateToSQ$accessor$FTak0Wwx(Unknown Source)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl$ByteBuddy$leA7obiG$auxiliary$pI0dPZUb.call(Unknown Source)
    at com.splwg.base.support.interception.BusinessMethodInterceptor.intercept(BusinessMethodInterceptor.java:71)
    at com.splwg.ccb.domain.rate.common.ApplyRateToSQ_Impl$ByteBuddy$leA7obiG.applyRateToSQ(Unknown Source)
    at
    com.splwg.ccb.domain.rate.calcRule.algorithm.applyRule.ServiceQuantityAlgComp_Impl.expandAndApplyRate(ServiceQuantityAlgComp_
    .java:239)
    at
    com.splwg.ccb.domain.rate.calcRule.algorithm.applyRule.ServiceQuantityAlgComp_Impl.processSeasonalSQ(ServiceQuantityAlgComp_
```

Logs

Expected Result: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

Actual Result: An error is encountered when billing for February month in a non-leap year with the seasonal period configuration.

Solution Design

- Update ***DeriveSQHelper*** to abstract the seasonal leap year logic into a common method, which includes validation to check if the seasonal start or end date is February 29th. If the seasonal year is not a leap year, adjust the date to February 28th.
- Invoke the common method from ***ApplyRateToSQ***, ***ApplyFlatCharge***, and ***ApplyLimitCharges*** for seasonal period validation, ensuring that all routines use the same logic.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	Y
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

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Java Changes

DeriveSQHelper_Impl (Existing)

Business Rules

- ….
- Introduce a new method validateAndCreateSeasonPeriod()
- If season data list is not null
  - If season start date and season end date is not null
    - If season start month is February and season start day is 29 and Bill Period Start Year is not a Leap Year
    - Then season start day = season start day - 1
    - If season end month is February and season end day is 29 and Bill Period Start Year is not a Leap Year
    - Then season end day = season end day – 1
- Build season start and season end dates using the reference start year
- If Season Start MMDD is less than Season End MMDD
  - If Derived Season End Date < Reference Date,
    - Add 1 year to Season Period
- If Season Start MMDD >= Season end MMDD (1201 – 0331) evaluate derived season end Date
  - If Season End Date < Reference Date,
    - Add 1 year to Season End Date
  - Else,
    - Subtract 1 year from Season Start Date
- Set the season Start and End Date into new Season Period
- …

ApplyRateToSQ_Impl (Existing)

Business Rules

- …
- Update updateBillCalculationLineValuePeriod() method
  - Create a new instance of class DeriveSQHelper
  - Create a new instance of class StartEndDateData
  - Invoke validateAndCreateSeasonPeriod()

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  - Save the season Start and End Date into billCalculationLineData
- …

ApplyFlatCharge_Impl (Existing)

Business Rules

- …
- Update buildSeasonalCalcLineData() method
  - Create a new instance of class DeriveSQHelper
  - Create a new instance of class StartEndDateData
  - Invoke validateAndCreateSeasonPeriod()
  - Save the season Start and End Date into billCalcLineData
- …

ApplyLimitCharges_Impl (Existing)

Business Rules

- …
- Update buildSeasonalCalcLineData() method
  - Create a new instance of class DeriveSQHelper
  - Create a new instance of class StartEndDateData
  - Invoke validateAndCreateSeasonPeriod()
  - Save the season Start and End Date into billCalcLineData
- …

Data Clean-Up / Fixes

N/A.

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Recommended Test Cases

Test Case 1: Verify the billing results for a non-leap year when the season end date is 02-29.

Test Case 1.1: Reported Issue

- Refer to Steps to Recreate.
- Generate a bill for period 02-01-2025 to 02-27-2025.

Expected Results: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

Test Case 1.2: Verify the results when Stepped Service Quantity Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Stepped Service Quantity** Calculation Rule with seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2025 to 02-27-2025.

Expected Results: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

Test Case 1.3: Verify the results when Flat Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Flat Charge** Calculation Rule with seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2025 to 02-27-2025.

Expected Results: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

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Test Case 1.4: Verify the results when Minimum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Minimum charge** Calculation Rule with seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2025 to 02-27-2025.

Expected Results: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

Test Case 1.5: Verify the results when Maximum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Maximum charge** Calculation Rule with seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2025 to 02-27-2025.

Expected Results: No error should occur when billing for February month in a non-leap year with the seasonal period configuration.

Test Case 2: Verify the results when season start is 02-29

- Refer to Steps to Recreate.
- Add a **Flat Charge** Calculation Rule with seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2025 to 03-29-2025.
- **Expected Results:** No error should occur when billing for March month in a non-leap year with seasonal start period as 02-29.

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Test Case 3: Verify the billing results for a leap year when the season end date is 02-29.

Test Case 3.1: Verify the results when Service Quantity Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Service Quantity** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2020 to 02-27-2020.

Expected Results: No error should occur when billing for February month in a leap year with the seasonal period configuration.

Test Case 3.2: Verify the results when Stepped Service Quantity Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Stepped Service Quantity** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2020 to 02-27-2020.

Expected Results: No error should occur when billing for February month in a leap year with the seasonal period configuration.

Test Case 3.3: Verify the results when Flat Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Flat Charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2020 to 02-27-2020.

Expected Results: No error should occur when billing for February month in a leap year with the seasonal period configuration.

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Test Case 3.4: Verify the results when Minimum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Minimum charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2020 to 02-27-2020.

Expected Results: No error should occur when billing for February month in a leap year with the seasonal period configuration.

Test Case 3.5: Verify the results when Maximum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Maximum charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-01 and Season End Date as 02-29.
- Generate a bill for period 02-01-2020 to 02-27-2020.

Expected Results: No error should occur when billing for February month in a leap year with the seasonal period configuration.

Test Case 4: Verify the billing results for a leap year when the season start date is 02-29.

Test Case 4.1: Verify the results when Service Quantity Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Service Quantity** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2020 to 03-29-2020.

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Expected Results: No error should occur when billing for March month in a leap year with the seasonal period configuration.

Test Case 4.2: Verify the results when Stepped Service Quantity Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Stepped Service Quantity** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2020 to 03-29-2020.

Expected Results: No error should occur when billing for March month in a leap year with the seasonal period configuration.

Test Case 4.3: Verify the results when Flat Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Flat Charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2020 to 03-29-2020.

Expected Results: No error should occur when billing for March month in a leap year with the seasonal period configuration.

Test Case 4.4: Verify the results when Minimum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Minimum charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2020 to 03-29-2020.

Expected Results: No error should occur when billing for March month in a leap year with the seasonal period configuration.

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Test Case 4.5: Verify the results when Maximum Charge Calculation Rule is configured

- Refer to Steps to Recreate.
- Add a **Maximum charge** Calculation Rule with leap year seasonal period configuration.
- Provide Season Start Date as 02-29 and Season End Date as 03-29.
- Generate a bill for period 02-27-2020 to 03-29-2020.

Expected Results: No error should occur when billing for March month in a leap year with the seasonal period configuration.

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